Nishal Stanislaus Silva, *PhD*

Machine Learning Engineer | Real-Time ML, Evaluation, LLM Systems | Production ML

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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Re: Senior Machine Learning Developer - Numa (Toronto, ON)

Dear Hiring Manager,

I am applying for the Senior Machine Learning Developer role at Numa. I am a machine learning engineer with a PhD from the University of Trento (2025) and 10+ years shipping ML systems, from real-time inference to evaluation pipelines. My most recent published system made real-time decisions in 14 ms on a Raspberry Pi 4 at F1 0.76, 74× faster than the DTW baseline it replaced (JAES 2026).

Here is how my work maps to the role:

- **Production ML, end to end.** On the EU-funded MUSMET project I owned the pipeline from acquisition to deployment, with <30 ms inference at >90% F1, reproducible pipelines and monitoring. At MAS Holdings I led deployed ML and vision systems for manufacturing at scale (up to 300% efficiency gain) and introduced CI/CD and code review.
- **Evaluation-first mindset.** I design baselines and ablations to separate real improvements from noise, most recently 6 methods over 100,000 scored trials (best ROC-AUC 0.822 vs. 0.599 baseline).
- **Real-time and orchestration.** Low-latency streaming inference is the core of my research, and I have built Airflow DAGs with S3-to-Snowflake pipelines in Docker. I use Claude Code daily for agentic development.

To be direct about the gaps: I have not built voice or SIP/WebRTC agents, and Ray Serve, Dagster, LiveKit and Vertex AI are new to me. My LLM and RAG work is tool-level rather than production-scale, and I have not worked at a startup in a product squad. My strength is evaluation, latency-constrained deployment and ownership from problem framing to monitoring, and I would ramp on the stack quickly.

I am based in Toronto, a Canadian permanent resident eligible to work in Canada without sponsorship, and available immediately. I would welcome the opportunity to discuss how I can help raise the evaluation and engineering bar at Numa.

Sincerely,

Nishal Stanislaus Silva, PhD